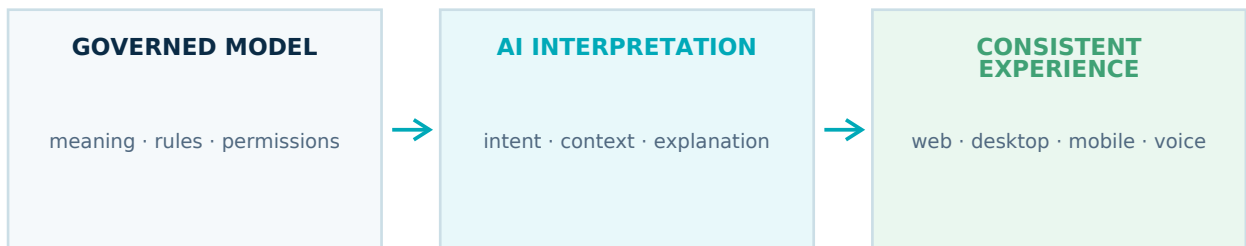


MODEL-FIRST

Model-First Enterprise Architecture

Rethinking Business Systems for the AI Era

“The primary artifact of a business system should be the governed business model - not the application code built around it.”



Model once. Govern explicitly. Interpret everywhere.

Davor Veljača

July 2026

AUTHOR'S NOTE

A strategic architecture vision

This paper explores a possible evolution of enterprise systems when AI can interpret a formal, governed business model.

It is not a technical specification for a single application and it does not depend on one AI vendor, one cloud provider or one user-interface technology. The central idea is deliberately technology-neutral: durable business meaning should belong to the organization, while the technologies that interpret it should remain replaceable.

DEPRO is used later as a case study to make the concept concrete. The purpose is to show how the same idea could affect day-to-day work, regulatory change, auditability, development roles and infrastructure strategy.

OVERVIEW

The proposition in one page

Traditional enterprise applications distribute business knowledge across database structures, procedures, services, user interfaces, workflow engines and integration code. This works, but over time the same rule can appear in multiple places and every change can propagate through a wide technical surface.

Model-First Enterprise Architecture changes the center of gravity. The governed business model becomes the primary artifact. The database stores both the model and operational data, and remains responsible for deterministic integrity, transactions, security and auditability. AI interprets user intent and the governed model, but it is not the source of truth.

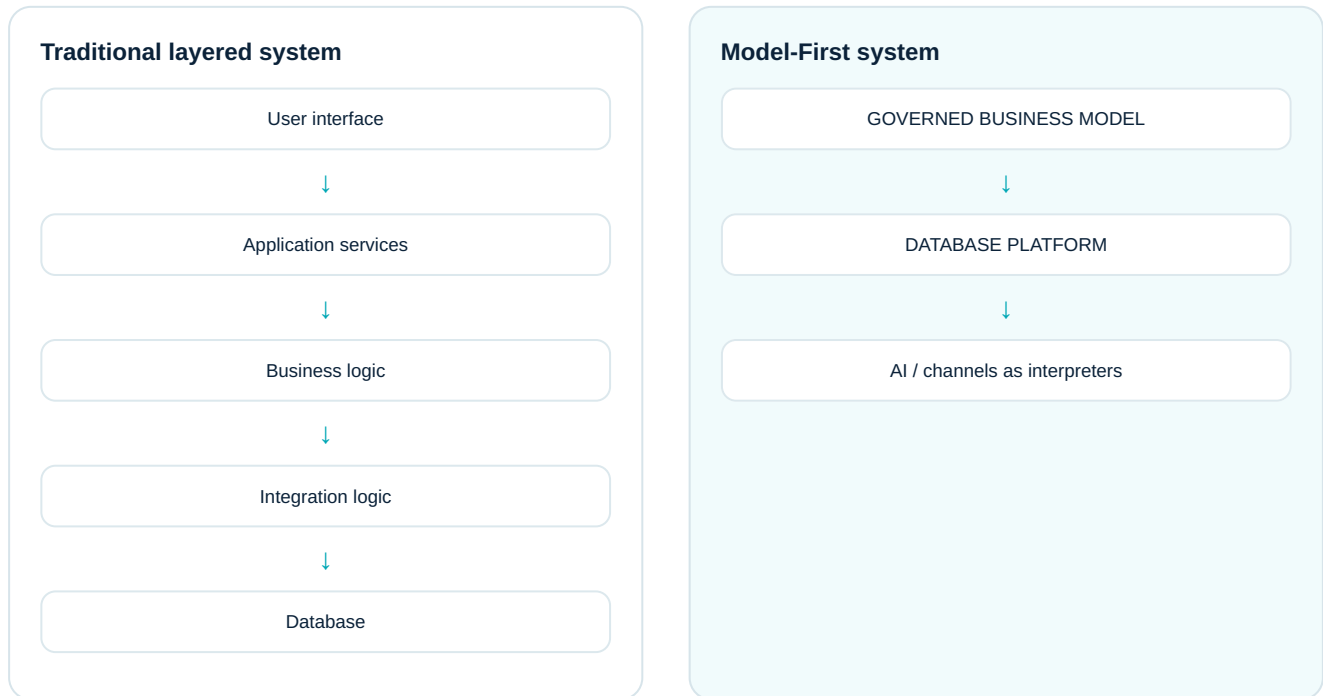
Five shifts

- Business change moves toward model change
- AI becomes interpreter, not owner
- UI becomes generated but uniform
- Audit links meaning to evidence
- Value shifts toward model, data and compute

CONTEXT

Why business systems may need a new paradigm

Current architectures were designed for a world in which computers had to be explicitly programmed for almost every interaction path.



The change is not simply fewer layers - it is one explicit home for business meaning.

AI changes one important assumption: technology can increasingly interpret intent, context and structured semantics. That does not eliminate the need for strict rules. It creates the possibility of moving durable business meaning out of repeated application implementations and into an explicit model that multiple channels can interpret consistently.

The issue is not that current architectures are wrong. The issue is that some of the constraints that shaped them are changing.

CORE PRINCIPLES

What Model-First means

- The governed business model is the single source of business meaning.
- The database is the final source of data integrity, transactions, authorization and deterministic enforcement.
- AI interprets; it does not own the rules and it does not bypass controls.
- Interfaces can be generated from intent and context, but are rendered through a stable design system.
- Model changes are versioned, approved, testable and auditable.
- Technology is replaceable. Business semantics and data remain durable organizational assets.

**Entities**

what exists

**Capabilities**

what can be achieved

**Rules**

what must be true

**Permissions**

who may do what

**Workflow**

how state may change

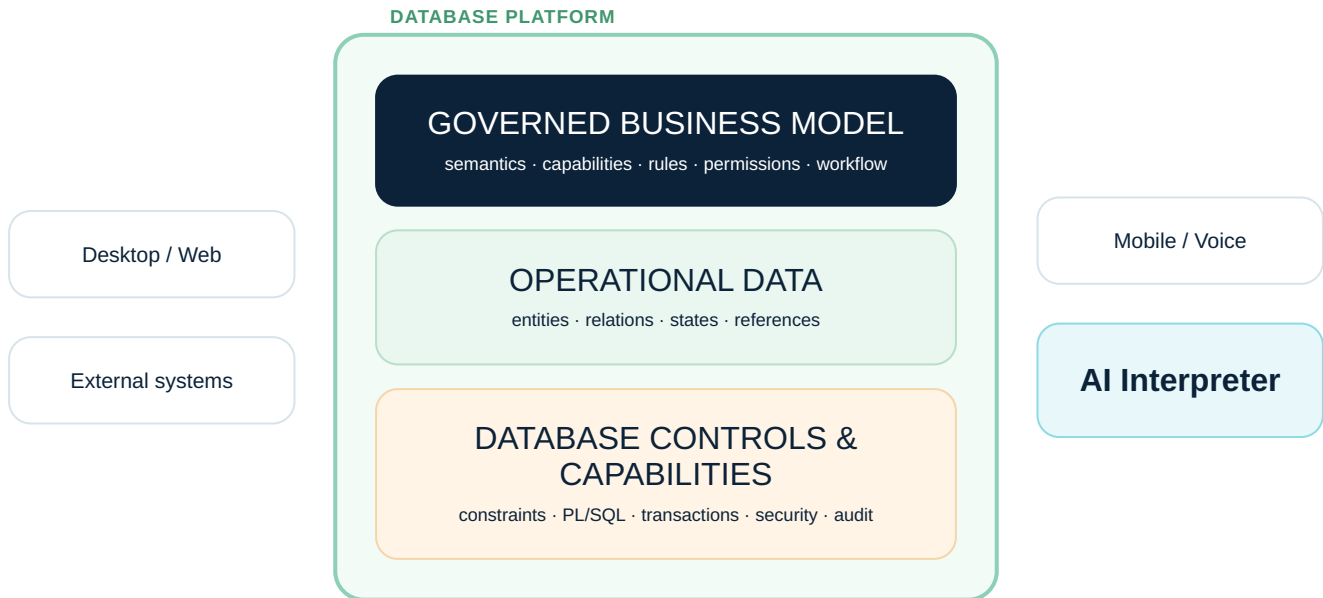
**Interaction**

what the user needs to understand

REFERENCE ARCHITECTURE

The model and the data live in the database platform

The architecture deliberately avoids introducing another business-logic middle tier that would become a new home for duplicated rules.



The model and the data live in the database; AI interprets them through governed operations.

Within the database platform, the semantic model and the operational data model coexist. The semantic model describes business meaning. Operational structures store durable facts. Constraints, authorization, transactions, audit mechanisms, functions and governed procedures enforce what must be deterministic.

AI reads only the authorized part of the model, guides the user and invokes only explicitly permitted database capabilities. It never receives unrestricted permission to invent its own SQL against production data.

THE MODEL

What the governed model describes

BUSINESS MEANING	Entities · relationships · vocabulary
GOVERNED BEHAVIOR	Capabilities · rules · permissions · workflow
INTERACTION SEMANTICS	Required information · explanations · channel intent

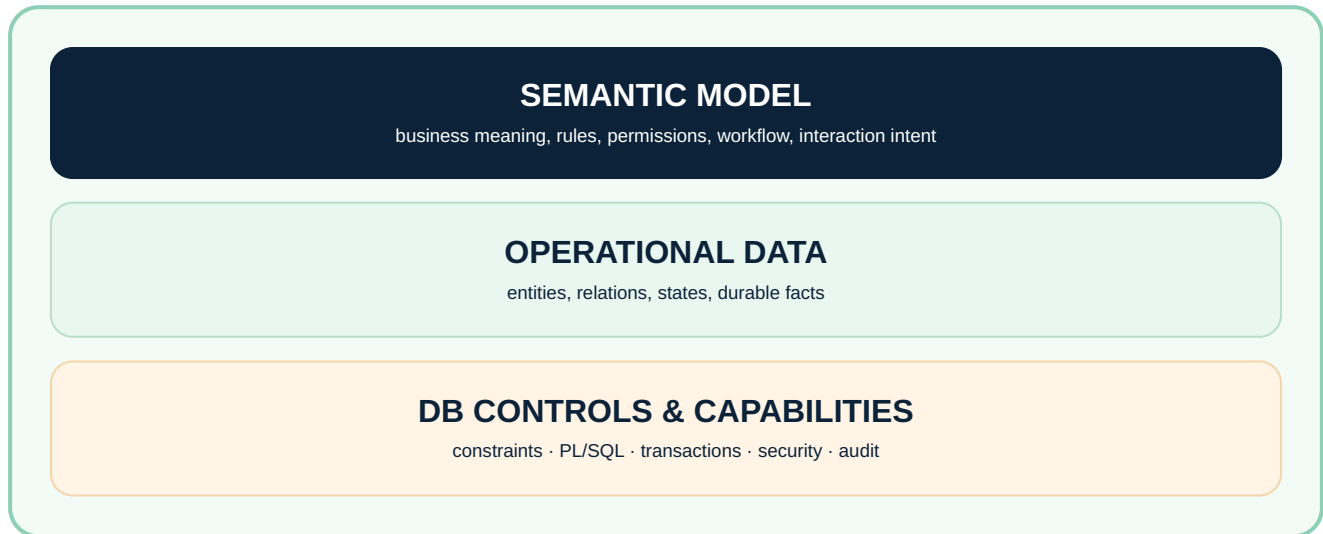
Intent → permitted operation → deterministic execution → evidence

The goal is not to store screens. The goal is to store a machine-interpretable description of business intent and constraints.

DATABASE

The database becomes more than storage

DATABASE PLATFORM



The database remains relational where that is the best representation of durable business facts. A concept such as Order 11 does not automatically require a physical table named ORDER_11. Physical structures should represent stable entities and relationships such as orders, order items, accounts, statuses, references and approvals.

Non-negotiable integrity belongs in deterministic database controls: data types, constraints, referential integrity, transaction boundaries, authorization, immutable audit records and governed functions or procedures.

Developers build reusable database capabilities rather than a new stack for every business request - for example validation, controlled state transition, persistence, audit recording and approved integration operations.

ARTIFICIAL INTELLIGENCE

Interpreter, not authority



Security boundary: private / sovereign cloud or on-prem; least privilege; controlled egress; no unrestricted production SQL.

AI is the adaptive layer between human intent and the formal system. It can conduct a conversation, identify missing information, explain a rule, assemble an authorized request and choose an appropriate visual presentation.

AI should not have unrestricted SQL access and should not become the final authority for accounting, regulatory or integrity decisions. Probabilistic interpretation can assist the user; deterministic database controls enforce what cannot be negotiated.

The AI layer may run in a private cloud, sovereign cloud or on-premise environment. The architectural requirement is isolation, least privilege, controlled data exposure and replaceability - not dependence on a specific provider.

USER EXPERIENCE

Generated, but intentionally uniform

Users should not receive a different visual language every time AI creates an interface.



The model describes required information, permitted actions and interaction intent. A fixed enterprise design system provides trusted components - forms, tables, approvals, warnings, timelines, dashboards and confirmation patterns. AI composes those components for the task and role; it does not invent a new design language on each interaction.

The result is adaptive without being chaotic: the content changes with context, while layout conventions and interaction patterns remain familiar.

OPERATING MODEL

How roles evolve

**Data / Model Architect**

semantics, rules, structures, versioning

**Developer / Database Engineer**

reusable DB capabilities, integrations, security, performance

**Business User**

validates real work and outcomes

**AI**

proposes, explains and interprets - never the source of truth

The largest shift is not the disappearance of engineering. It is the reduction of repeated implementation work.

Data / Model Architects own formal semantics, data structures and the definition of governed business rules. Developers and database engineers implement reusable deterministic capabilities, PL/SQL where appropriate, integrations, performance, security, tooling, migration and platform evolution. Business users validate whether the model actually represents real work.

AI can generate code, tests and proposals, but high-impact changes remain governed, reviewed and traceable.

GOVERNANCE

Auditability and compliance by design



From business meaning to technical evidence.

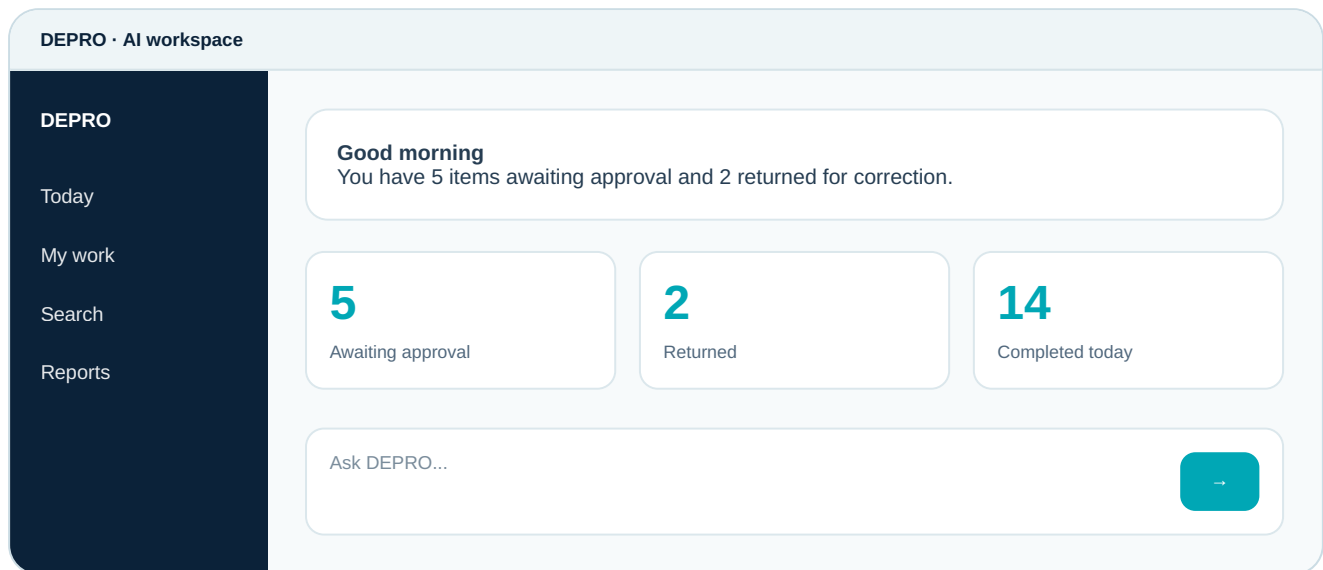
Every important action should be reconstructable. Evidence should identify who initiated it, which model version applied, which rules were evaluated, which governed capability executed, what state changed and what result followed.

The audit trail is not merely a technical log. It links business meaning to deterministic execution evidence.

AI may explain that evidence in natural language, but the underlying evidence must remain immutable, deterministic and independently verifiable.

CASE STUDY

DEPRO 2030 - the workday starts with intent, not menus



DEPRO is used here to demonstrate the experience of a Model-First system without turning the architecture into a DEPRO-specific solution.

The user begins with obligations and intent: items awaiting approval, items returned for correction, exceptions and deadlines. Conversation, structured forms, reports and approvals share the same model and the same security context.

The system does not require the user to remember where a function is located. The user expresses what needs to be achieved; the model determines what is allowed and what information is required.

CASE STUDY

Order 11 as a model-driven operation

AI guidance

I want to open Order 11.

The approval account is missing.

Use the default settlement account.

Done. One additional approval is required because of the amount.

Order 11

Order number

2400-001

Debit account

SETTLEMENT-01

Credit account

CONTROL-001

Amount

125,000.00 EUR

Additional approval required

Submit for approval

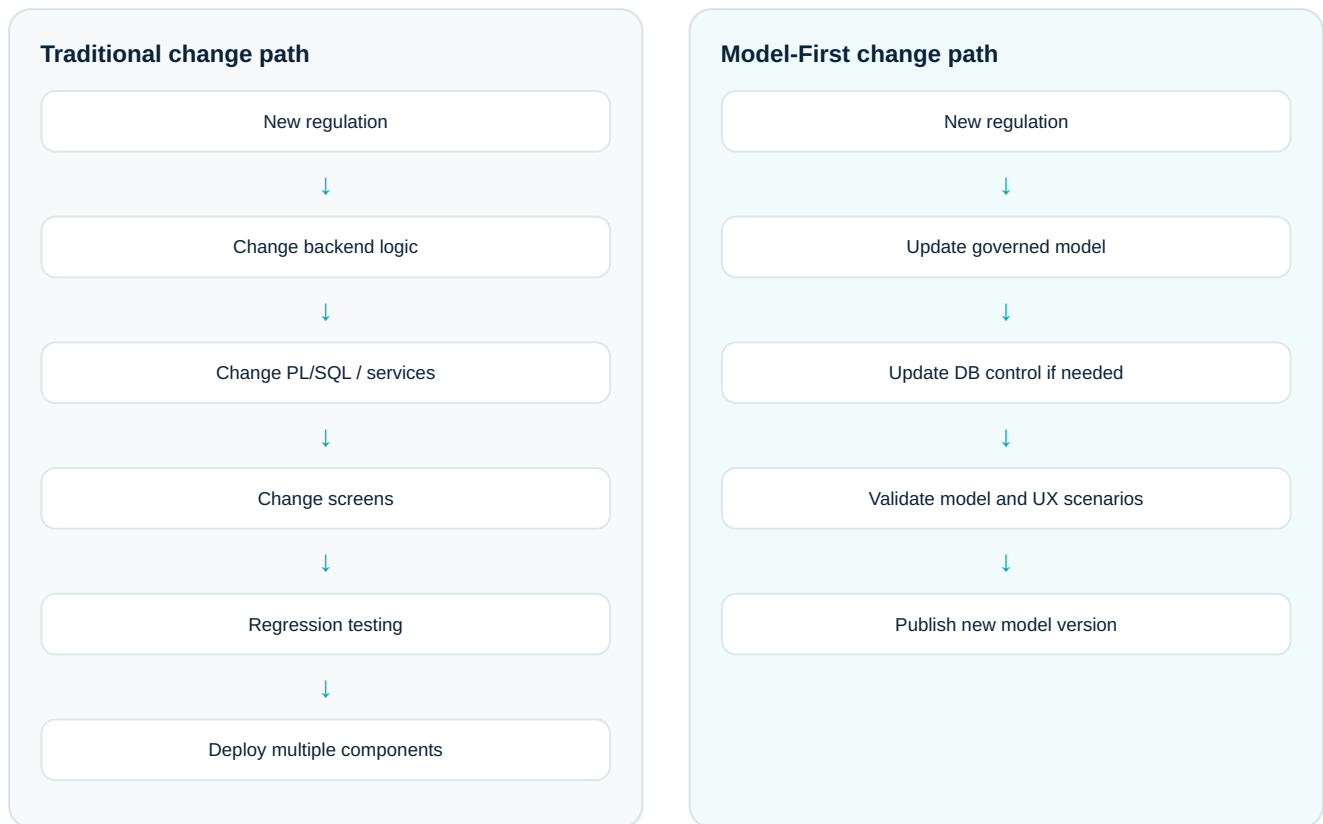
The form is not the primary artifact. The governed model defines required information, allowed values, validation rules, approval conditions and permitted operations.

The user may provide data conversationally or directly in a structured workspace. Both modes operate on the same context and the same deterministic controls.

When the user is ready to submit, AI invokes a governed database capability. The database validates and executes. AI does not override a failed rule.

CASE STUDY

A regulatory change without rebuilding the application



One business change - two very different change surfaces.

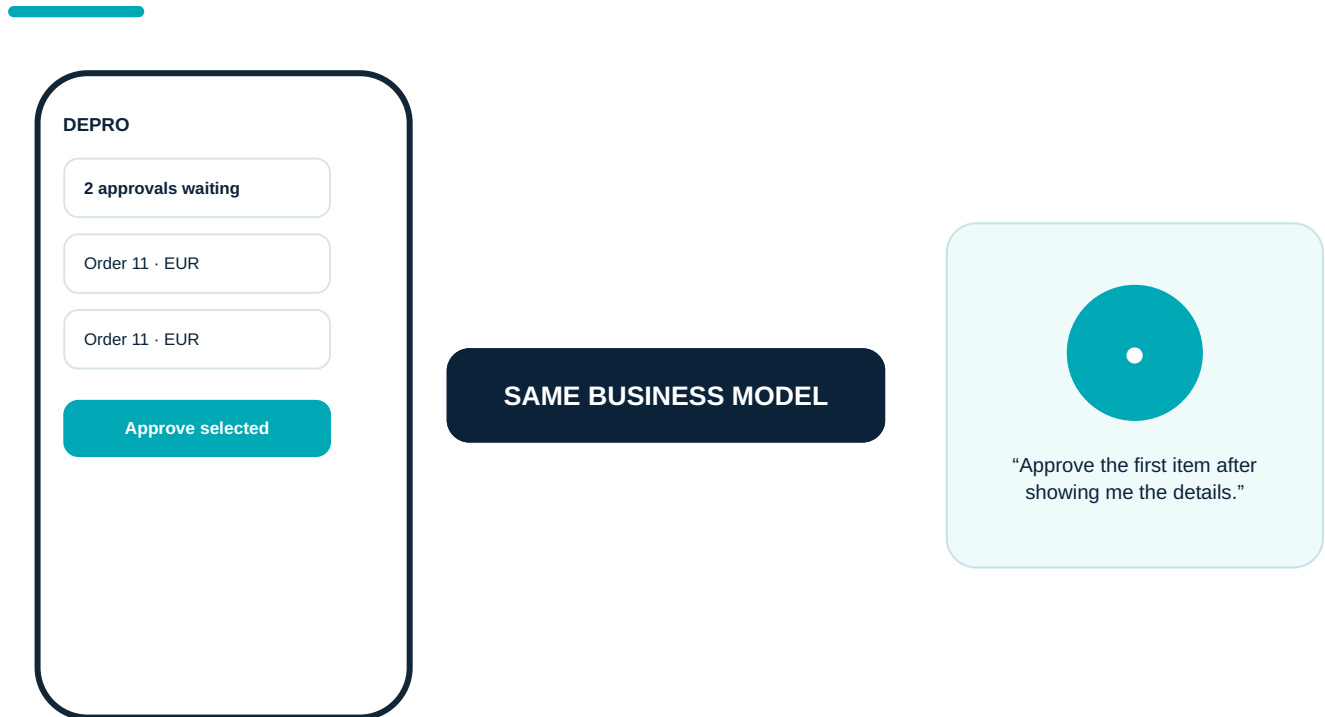
Consider a new regulation that changes a threshold, introduces an additional approval condition or requires new evidence.

In a mature Model-First system, the primary change is made to the governed model and, where necessary, to deterministic database controls. All channels interpret the same published model version.

Testing remains essential. What changes is the testing surface: model correctness, enforcement, generated interaction, integration effects and user acceptance - instead of repeatedly changing the same rule across multiple application layers.

CASE STUDY

The channel changes. The business model does not.



Same permissions, rules and audit - different presentation by channel.

Desktop, web, mobile and voice become different expressions of the same governed semantics.

The amount of information shown and the interaction pattern can change by device and task, while permissions, rules, lifecycle and audit requirements remain identical.

A mobile user may approve an item in a compact flow; a desktop controller may see a detailed table; a voice interaction may request confirmation. All three act on the same modeled capability.

CASE STUDY

Audit replay - reconstructing a DEPRO decision



Why was this Order 11 allowed? The system shows the business reason and the execution evidence.

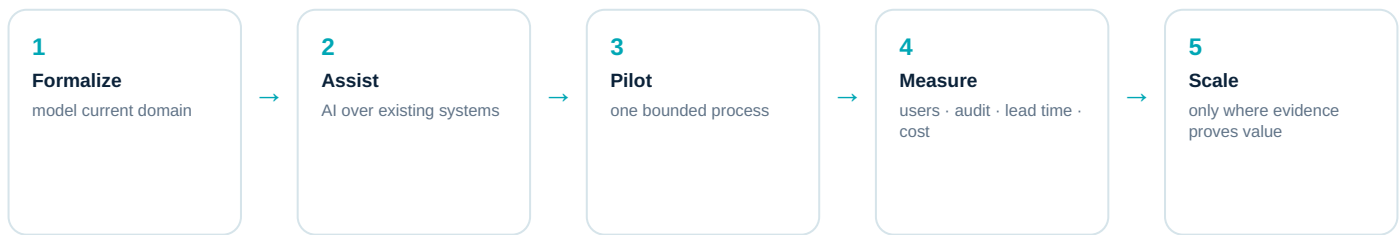
A historical transaction should be replayable as a chain of evidence, not merely searchable as a log entry.

Example: the reviewer selects an Order 11 from a prior period and sees the initiating user, the AI-resolved intent, the active model version, the exact rule that applied, the database validation result, approvals, state changes and final outcome.

The key distinction from the general audit principle is practical: the reviewer can reconstruct why this specific transaction was allowed or rejected, in business language, while the evidence remains backed by immutable technical records.

ADOPTION

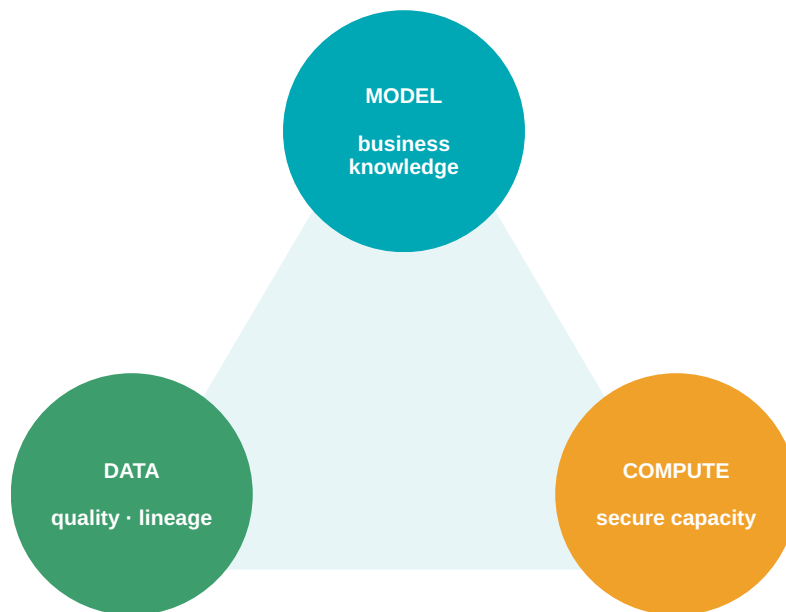
An evolutionary path, not a disruptive replacement



1. Formalize the current domain without changing the user experience.
2. Introduce AI as a constrained assistant over existing systems.
3. Pilot one bounded process as a true Model-First implementation.
4. Measure user acceptance, auditability, lead time, defects, stability and total cost.
5. Scale only where evidence shows the model-driven approach is simpler, safer and easier to change.

STRATEGIC CONSEQUENCES

Where enterprise value may shift



The combination matters more than any single component.

If more business change can be expressed through a governed model instead of bespoke application development, the marginal cost of software customization can decline.

At the same time, value moves toward three durable assets: the quality of the business model, the quality and governance of data, and secure computational capacity.

This is a rebalancing, not a promise that IT becomes cheap. Less repetitive application code may coexist with greater investment in data architecture, AI capability, security and compute.

Why compute capacity becomes more strategic

>2x

Projected growth in global data-centre electricity demand by 2030 (IEA base case).

External context

The IEA projects around 945 TWh of global data-centre electricity consumption by 2030, with AI the largest driver of growth.

AI makes many more forms of computation economically useful: continuous interaction, retrieval, simulation, validation, analytics and private inference. Even as hardware becomes more efficient per unit of performance, total demand can grow because the number and intensity of AI-driven workloads grows.

The IEA projects global data-centre electricity demand to more than double by 2030, with AI as the most important driver of growth. This does not prove that every organization needs its own data center; it does show that compute, power and capacity planning are becoming structural strategic questions. [1]

For Model-First systems, this matters because the business model is only useful if the organization can interpret it securely, reliably and at the required scale.

INFRASTRUCTURE STRATEGY

Build, rent or hybrid

CLOUD / RENT	OWNED / BUILD	HYBRID
• Elastic capacity • Fast access to new AI services • Lower initial capital • Greater provider dependency	• Control and sovereignty • Predictable capacity • Potential economics at high utilization • Capital and operational complexity	• Private baseline + elastic peaks • Workload placement by sensitivity • Gradual expansion • Requires strong architecture discipline

Cloud / rent: strongest when elasticity, speed of adoption and access to rapidly changing AI services matter most.

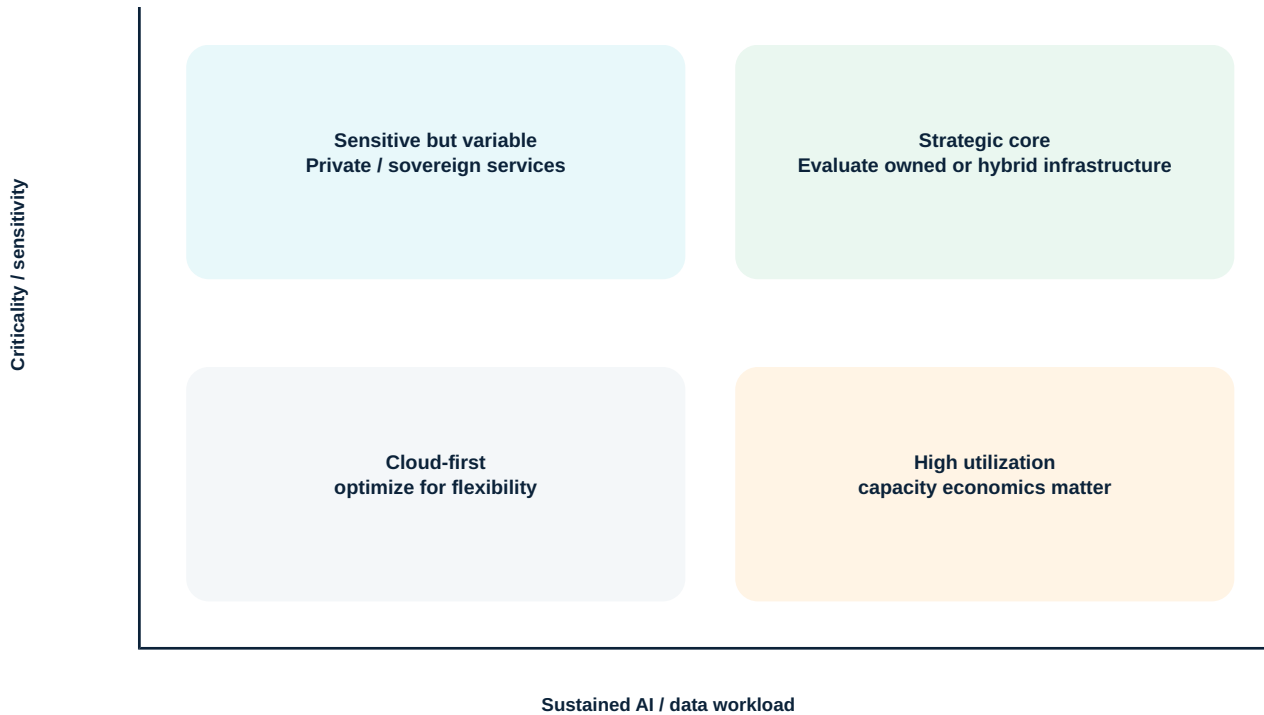
Owned / build: strongest when workloads are sustained, sensitive, predictable and strategically critical - and when the organization can operate infrastructure efficiently.

Hybrid: strongest when a stable private baseline can be combined with external elasticity and specialized services.

The correct answer is workload-specific. The objective is optionality: avoid premature overinvestment, but also avoid dependencies that later become difficult or expensive to reverse.

INFRASTRUCTURE STRATEGY

When should greater infrastructure control be considered?



The case becomes stronger when three conditions converge: sustained high utilization, high sensitivity or sovereignty requirements, and operational criticality.

For a large European group, the practical step is not to declare “build a data center” as a universal answer. It is to build multi-year demand scenarios now - AI inference growth, power and cooling, security, resilience, accelerator refresh cycles, staffing, sovereignty and exit costs - so that infrastructure choices are made before capacity becomes a bottleneck.

This direction is consistent with the broader European focus on sovereign AI and cloud capacity. In June 2026, the European Commission proposed the Cloud and AI Development Act with an objective to substantially expand EU data-centre capacity. [2][3]

ECONOMIC HYPOTHESIS

Cheaper customization, more valuable infrastructure



The Model-First thesis suggests a rebalancing rather than a simple cost reduction.

Software value shifts from the volume of custom code toward the quality of formal business semantics. Engineering effort shifts from one-off functionality toward reusable database capabilities, security, integration and platform quality. Infrastructure planning becomes more strategic as AI usage becomes continuous and embedded.

A strong long-term position may be a deliberate combination of owned capabilities and external elasticity rather than an ideological commitment to either cloud or on-premise.

GUARDRAILS

What does not change

- Accountability remains with the organization and its authorized people.
- Security, segregation of duties and regulatory controls remain mandatory.
- Critical integrity rules remain deterministic.
- Data quality, lineage and retention remain fundamental.
- Users still need stable, predictable and accessible experiences.
- Testing remains essential - but increasingly tests the model, enforcement and interpretation rather than duplicated application logic.

GUARDRAILS

What this architecture is not

- It is not unrestricted AI access to production tables.
- It is not a claim that natural language should replace every visual interface.
- It is not a claim that a database alone solves every distributed-systems or integration problem.
- It is not an argument that developers or engineering discipline become unnecessary.
- It is not a mandate that every organization build its own data center.
- It is a proposal to make durable business meaning explicit, governed and reusable across technologies.

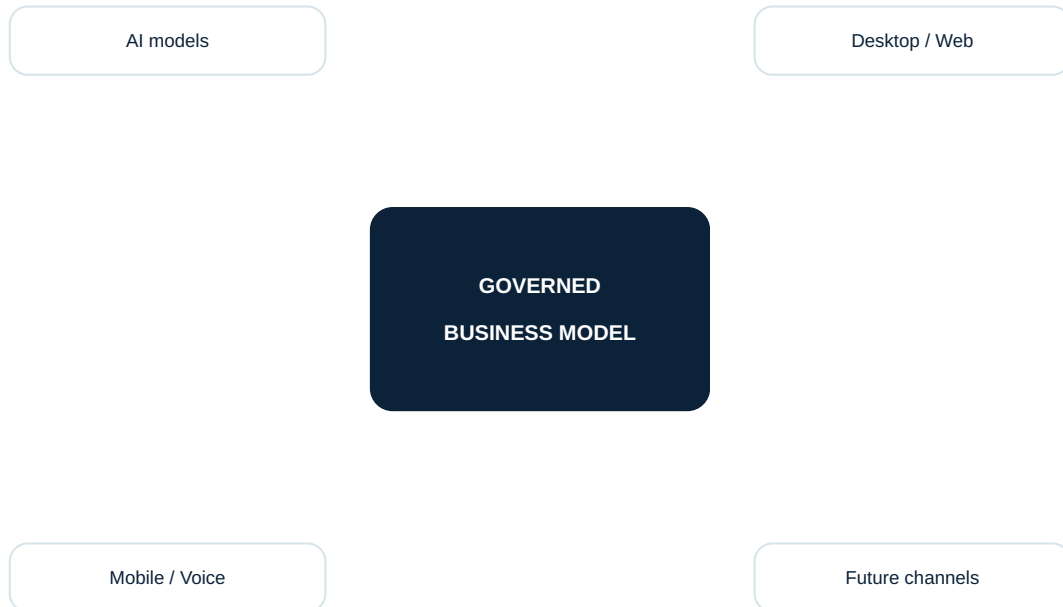
RESEARCH AGENDA

Open questions that should be tested, not ignored

- How should the modeling language balance human readability with machine precision?
- Which rules belong in semantic metadata and which must always be deterministic database controls?
- How should model versions be approved, tested, certified and rolled back?
- How can AI interpretation be made sufficiently reproducible for high-risk processes?
- How should generated interaction be constrained by design-system and accessibility standards?
- How should organizations measure the trade-off between lower customization cost and higher compute consumption?
- Which workloads justify strategic control of infrastructure, and which should remain elastic and external?

VISION 2035

Technologies will change. The governed business model should remain.



Model once. Technologies change. Interpretation adapts.

The durable asset is the organization's explicit understanding of how its business works: semantics, rules, permissions, processes and data.

If that understanding is precise enough for people to govern and structured enough for machines to interpret, enterprise systems can become less about repeatedly programming applications and more about continuously modeling the business.

The AI model, user channels and infrastructure can change. The business model remains the durable center.

CLOSING THOUGHT

A question worth asking

If we were designing enterprise systems from first principles today - with modern databases, governed data and AI capable of interpreting structured semantics - would we still build them exactly as we did twenty years ago?

Perhaps the next generation of enterprise software will not be defined by how much code we can generate, but by how precisely we can model the business.

Model once. Govern explicitly. Interpret everywhere.

REFERENCES

Selected external context

[1] International Energy Agency (IEA), Energy and AI, 2025. The report projects global data-centre electricity consumption to more than double to around 945 TWh by 2030, with AI the largest driver of growth.

[2] European Commission, Cloud and AI Development Act, proposal announced June 2026. The initiative aims to strengthen European cloud and AI capacity and significantly expand data-centre capacity.

[3] European Commission, Strengthening Europe's Tech Sovereignty, 2026. The policy framing emphasizes Europe's ability to develop and control key technologies, data and infrastructure.

[4] OECD, Competition in Artificial Intelligence Infrastructure, 2025. The report describes advanced chips, data centres, power, cooling and networking as core physical inputs to AI infrastructure.

These references provide external context for the infrastructure chapter. The Model-First architecture itself is an original conceptual proposal and should be validated through bounded pilots and empirical measurement.